Lumber & purchasing basis

Prepared 2026-09-08. Based on the current Python model: 48 × 96 in inside; 54.37 × 102.37 × 84 in outside. Screw assembly, three interior light supports, slab-supported walls and folded floor liner.

Order quantities include the allowances stated below. They are not a fastening schedule or a structural approval. Prices are intentionally blank in the companion CSV for your supplier quotes. No floor deck, platform joists, floor insulation, slab anchors or nails are on this list.

Qty	Stock / item	Basis and selection notes
28	2x3 × 8 ft lumber	25 allocated to modeled cuts + 3 allowance boards for backing/spares.
4	2x3 × 10 ft lumber	4 allocated to modeled cuts Long plates exceed 8 ft; do not substitute short stock without a splice detail.
8	2x4 × 8 ft lumber	6 allocated to modeled cuts + 2 allowance boards (one seam-backing allowance, one spare). Grade/species and roof load check pending.
1	4x4 curb offcut	One piece at least 30 in long; machine to 28 × 2.810 × 3.375 in. Finalize curb detail first.
4	1x3 × 8 ft furring	Finish/perimeter support allowance only; actual profile and attachment follow panel requirements.

Select straight, dry framing lumber. Confirm availability of 10-foot 2x3 stock with the lumberyard. Long bottom plates need moisture-compatible wood selection and a capillary break. Changing lumber dimensions or finish thickness changes the enclosure geometry.

Roof member size is provisional: total light and hanging-hardware weight has not been supplied. Do not purchase a connector package or claim a roof capacity from these quantities alone.

Sheets, insulation & floor liner

Sheet counts come from a conservative, kerf-aware 48 × 96 in cutting layout, not net area divided by sheet area. Stock-sheet layouts are included as CSV coordinates.

Qty	Sheet material	Basis / confirm before ordering
9	0.06 in wall/ceiling liner, 4 × 8 ft	191.1 ft² net; 288 ft² purchased. Select compatible product and verify support/finish requirements.
8	0.375 in plywood, 4 × 8 ft	172.2 ft² net; 256 ft² purchased. Verify bracing or roof span rating.
2	0.5 in plywood, 4 × 8 ft	38.7 ft² net; 64 ft² purchased. Verify bracing or roof span rating.
8	0.25 in finish substrate, 4 × 8 ft	159.7 ft² net; 256 ft² purchased. Select compatible product and verify support/finish requirements.
1	0.5 in finish substrate, 4 × 8 ft	32.0 ft² net; 32 ft² purchased. Select compatible product and verify support/finish requirements.

Qty	Insulation / floor	Basis / selection notes
Offcut	1 in header pack	0.8 ft² net; use compatible insulation offcut.
170 ft²	2.5 in cavity insulation	141.2 ft² net + 15%, rounded up. Actual bag count and cavity fit depend on selected product.
40 ft²	3.5 in cavity insulation	33.1 ft² net + 15%, rounded up. Actual bag count and cavity fit depend on selected product.
1	6 × 10 ft waterproof plastic liner	One continuous sheet; model assumes 40 mil. Includes fold/curb allowance; verify template, membrane suitability and corners before cutting.

Floor liner and wall/ceiling liner are separate systems. Do not buy multiple floor strips from the geometric cut list.
The membrane must remain continuous around folded corners and over the supported doorway curb. Finish liner thickness is schematic; match actual material and adhesive requirements.

Sealing, door & screw supplies

Allowance quantities below support budgeting and supplier quotes. Final screw diameters, lengths, spacing and connector models still require a joint-by-joint schedule.

Qty	Item	Selection / allowance
1 roll	Sill gasket, at least 3 in wide	At least 31 linear ft; thin capillary break, not anchorage.
40 ft²	Membrane protection layer	Compatible underlayment / equipment-foot protection; choose for slab roughness and liner.
20 ft	Door compression gasket	Compatible with the finished jamb, latch and threshold.
180 ft	Compatible seam tape / joint system	Planning allowance for panel, perimeter and door transitions; use approved welded/bonded joints if required instead.
6 tubes	Compatible sealant, approx. 10 oz	Allowance only; substrate, joint width and membrane chemistry determine actual usage.
220 ft²	Liner adhesive coverage	Only if the selected liner requires adhesive; divide by manufacturer coverage for container quantity.
3	Door hinges	Provisional count; select for actual leaf weight, size and screw holding.
1 set	Compression latch and handles	Operable from inside; match gasket compression and door thickness.
1 set	Jamb/stop stock and threshold seals	Final profile and lengths follow the finished door detail; not fully modeled.

All membrane fastening stays above the tray rim. Seal ceiling penetrations for hanging hardware. Select corrosion protection suitable for humid service and any treated lumber. These supplies do not replace active dehumidification or outdoor exhaust.

Fasteners & items to finalize

A drill/driver-based build is intended. Structural wood screws and metal-connector screws are different products; do not substitute generic deck or drywall screws for specified structural fasteners.

Qty	Item	Use / decision required
250	Structural wood screws	Purchase allowance for framing/corners. Size, length and end/edge distances TBD; use load-rated products for the selected joints.
500	Plywood bracing/roof screws	Allowance only. Select screws with a verified sheathing attachment schedule; do not copy a nail schedule.
250	Interior substrate screws	Allowance only; choose for the selected panel and support system. Avoid overdriving or penetrating the floor membrane.
10	Roof ties / angles	One per member end as a planning allowance. Exact screw-installable connector, count and orientation need load/geometry verification.
8	Wall-corner ties / angles	Two per corner as a planning allowance; not a verified bracing or stability solution.
TBD	Connector screws	Use only the exact screws and hole counts approved for each chosen metal connector. Quantity follows connector selection.
TBD	Rated light-hanging kits	Quantity follows light count and manufacturer suspension points. Confirm total lights + drivers + cables + hanger weight.
1 set	Tools / consumables	Drill/driver, correct bits, clamps, tape, square, level and cutting tools; check what you already own.

Not priced or sized: grow lights, drivers, fans, dehumidifier, ducts, controls, power distribution, service sleeves, condensate collection and grow equipment. Choose these after operating conditions, electrical needs and light loads are known.

Lumber cut reference

Dimensions below are model dimensions in inches, before field fitting. Nominal lumber names differ from actual section sizes. Do not round all decimal cuts down; finalize the joinery and measure actual stock first.

Qty	Part	Actual section (in)	Cut length (in)
2	Bottom plate	1.5 × 2.5	12.81
1	Bottom plate	1.5 × 2.5	53.62
2	Bottom plate	1.5 × 2.5	96.62
2	Door jack stud	1.5 × 2.5	69.915
2	Door rail	1.5 × 2.5	24.75
2	Door stile	1.5 × 2.5	67.75
3	Header cripple	1.5 × 2.5	3.46
2	Top plate	1.5 × 2.5	53.62
2	Top plate	1.5 × 2.5	96.62
18	Wall stud	1.5 × 2.5	76.875
1	Door header	1.5 × 3.5	31
3	Light-support crossmember	1.5 × 3.5	53.62
2	Roof end member	1.5 × 3.5	53.62
4	Roof side rim segment	1.5 × 3.5	24.78
4	Roof side rim segment	1.5 × 3.5	22.28
1	Doorway curb backing	2.81 × 3.375	28

The curb is a shaped 4x4 offcut, not standard 2x stock. Corner backing, sheet-seam blocks, finish furring and door stops are allowance materials, not included in the modeled cut lengths above.

lumber_stock_plan.csv assigns modeled cuts to each board, allowing 1/4 in total end trim and 1/8 in kerf per cut. Purchase extras are outside that allocation. sheet_stock_plan.csv shows every panel piece on its allocated stock sheet, without rotation.

Before ordering / source notes

Use this page to resolve the remaining product choices with your supplier or builder. The model has no verified hanging-load capacity, fastening schedule or freestanding stability calculation.

LIGHT LOAD: Enter total suspended weight in config.json. Confirm roof lumber grade/species, connections, bearing and wall stability before treating the 2x4 members as adequate.

PANEL SUPPORT: Walls are framed at up to 24 in; roof support spacing reaches 26.28 in. Verify the actual plywood/substrate span rating, orientation and support requirements. Thin finish substrate may require additional furring.

SHEET JOINTS: Wall panel splits use modeled stud centerlines. Roof-cap strips require a backed longitudinal seam; one spare 2x4 is allowed for that backing in the default design. Support ceiling perimeter and all required edges. Actual sheet sizes, damaged edges and installation gaps may change cuts.

WATERPROOFING: Confirm actual floor membrane, corner method, joint chemistry, adhesive coverage and liner blank template. The default 6x10 ft floor sheet includes allowance, but verify the doorway wrapping and corner folds before cutting.

DOOR: Finalize leaf fit, jamb stops, curb/threshold, hinges and inside-operable latch. The modeled door blank is provisional and influences sheet purchases.

STOCK QUANTITIES: Sheet counts reflect one feasible conservative cutting layout, not an optimum. Different approved seam layouts may save material. No additional full spare sheets are included; inspect stock and consider returns policy.

Technical references (general guidance; none certifies this enclosure):

Simpson Strong-Tie: fastener installation guide

Simpson Strong-Tie: approved connector screws

American Wood Council: loads and span-table tutorial

DOE: sill plate air sealing